

Question

A global asset manager deploys an LLM-based portfolio commentary generator. After release, journalists discover that the model occasionally invents fictitious analyst opinions attributed to real people at the firm. The firm's legal team, PR team, and regulators all become involved. Which combination of risk categories from the GARP taxonomy does this scenario most directly implicate?

- A. Individual risk and operational risk only.
- B. Organizational risk (reputational and legal) and societal risk (misinformation at scale).
- C. Societal risk only.
- D. Strategic risk only.

Answer: B

Explanation

Reputational damage, legal liability, and regulatory scrutiny are organizational risks. The generation of misleading synthetic content attributed to real people amplifies misinformation risk at the societal level. Option A understates the scope. Option C misses the firm-level harm. Option D is too narrow.

Question

A bank's automated credit-approval system generates a recommended decision for every application. The bank has decided that for applications with a predicted default probability between 15% and 25% (an uncertain band), the model's recommendation should be reviewed by a human underwriter. What is the most appropriate rationale for this design choice?

- A. To reduce compute costs.
- B. To ensure meaningful human oversight in a zone of genuine uncertainty, catching model errors and exercising judgment where the model is not confident.
- C. Because the regulator requires it for all applications.
- D. Because humans are always more accurate than models.

Answer: B

Explanation

Human-in-the-loop review is best used in zones of genuine model uncertainty, where the marginal value of human judgment is greatest. Option A is a secondary benefit. Option C is over-broad. Option D is an overstatement.

Question

A quant team needs to build a daily volume forecast for a large-cap equity. Two approaches are proposed: (i) a traditional ARIMA model, and (ii) a recurrent neural network (LSTM) trained on high-frequency order-book data. The team has 8 years of minute-level data. Which approach is most likely to be appropriate, and why?

- A. ARIMA, because neural networks cannot handle sequential data.
- B. LSTM, because it can capture complex non-linear dependencies in high-frequency data that linear time-series models may miss, provided sufficient data and computational resources.
- C. ARIMA, because it is always more accurate for financial time series.
- D. Neither approach; only tree-based models apply.

Answer: B

Explanation

LSTMs are specifically designed for sequential data and can capture non-linear patterns in high-frequency series. ARIMA remains useful but is linear and limited. Option A is wrong. Option C is not generally true. Option D is incorrect.

Question

A risk analyst generates a scatter plot of trading volumes versus trade sizes for a broker's equity order flow. A small cluster of orders shows unusually high volume with unusually small trade size, far from the main cloud of observations. What might this cluster indicate?

- A. Normal market activity.
- B. A potential anomaly or pattern of concern, such as algorithmic order splitting or spoofing behaviour, warranting further investigation.
- C. A data-entry error with certainty.
- D. Nothing of interest.

Answer: B

Explanation

Visual inspection of features can reveal clusters or outliers that indicate anomalies worth investigating. Option A ignores the deviation. Option C is premature. Option D misses the signal.

Question

A junior analyst decides to remove all observations with any missing value from a dataset used for a credit-scoring model. The dataset originally had 100,000 rows; after the removal, only 62,000 remain. The senior data scientist raises a concern. What is the most likely concern?

- A. The dataset is too small.
- B. Listwise deletion can introduce bias if missingness is systematic, distorting the model's representation of the underlying population.
- C. The dataset should be expanded.
- D. The dataset cannot be used.

Answer: B

Explanation

Listwise deletion is problematic when missingness is non-random. Biases can be introduced if missingness correlates with the target or other important features. Option A ignores the deeper issue. Options C and D are secondary or incorrect.

Question

A credit risk team considers adding a new feature to a default model: “number of missed payments in the last 6 months.” The team debates whether this feature should be used at training and prediction time. Which is the best approach?

- A. Use it at training only.
- B. Use it at both training and prediction if it is available at prediction time in real-world deployment; otherwise it introduces leakage.
- C. Use it at prediction only.
- D. Do not use it at all.

Answer: B

Explanation

Features must be available at both training and prediction time. Features computed using future information cause leakage. Options A, C, and D are incorrect.

Question

A team fits a logistic regression with L2 penalty to a credit-default dataset with 5,000 observations and 200 features. They observe that increasing the regularization strength decreases both training and test accuracy. What does this suggest?

- A. The regularization strength should be further increased.
- B. The model may be underfitting; regularization strength is too high, or the model needs more complexity/features.
- C. Regularization is not needed.
- D. The target variable is wrong.

Answer: B

Explanation

If increasing regularization decreases both training and test performance, the model is being over-penalised (bias dominates). The right regularization strength balances bias and variance. Options A, C, and D are not supported by the symptom.

Question

An analyst applies PCA to 20 correlated macro features and retains 4 components capturing 92% of variance. The resulting model performs well but the business team complains that the components are hard to interpret. Which trade-off is at play?

- A. Accuracy vs cost.
- B. Dimensionality reduction and stability vs interpretability — components are linear combinations of features, obscuring business meaning.
- C. Overfitting vs underfitting.
- D. Randomness vs determinism.

Answer: B

Explanation

PCA improves dimensionality and stability but sacrifices interpretability, since components do not correspond to individual business features. Options A, C, and D are not the primary trade-off.

Question

A random forest model ranks “age” as the most important feature for predicting fraud. A careful analyst notices that “age” and “retirementstatus” are highly correlated. What is a known limitation of feature importance in this context?

- A. Feature importance is always wrong.
- B. Feature importance can be inflated for correlated features; the model may arbitrarily select one and mask the importance of the correlated sibling.
- C. Feature importance is only valid for linear models.
- D. Feature importance is a fairness metric.

Answer: B

Explanation

Impurity-based feature importance is biased toward correlated features. Permutation importance or SHAP is often more reliable in the presence of correlation. Options A, C, and D are incorrect.

Question

A model uses log-transformed revenue as a feature. The coefficient on $\log(\text{revenue})$ is 0.30. How should this coefficient be interpreted?

- A. A one-unit increase in revenue increases the target by 0.30.
- B. A 1% increase in revenue increases the target by approximately 0.30% (if target is also in logs) or by 0.003 units (if target is in levels), depending on the model specification.
- C. The coefficient is meaningless.
- D. Log transformation destroys interpretability.

Answer: B

Explanation

In log-log models, coefficients are elasticities. In log-linear models, coefficients are semi-elasticities. Option A misses the log scale. Options C and D are incorrect.

Question

A deep neural network with 10 million parameters achieves 99.5% training accuracy and 68% test accuracy on a labelled financial dataset of 100,000 observations. What should the team do?

- A. Increase model size.
- B. Reduce model complexity, add regularization (dropout, weight decay), gather more data, or use early stopping — the model is overfitting.
- C. Use a simpler loss function.
- D. Change the target variable.

Answer: B

Explanation

The huge gap between training and test accuracy signals overfitting. Standard remedies: reduce complexity, regularize, gather more data. Options A, C, and D do not address overfitting.

Question

In a regulated credit-scoring context, a bank must explain individual decisions to customers and regulators. The bank's data science team proposes a highly accurate gradient-boosted model. Which combination of strategies best satisfies both accuracy and explainability needs?

- A. Use the model as-is without explanation.
- B. Use the model with post-hoc explainability (SHAP, counterfactuals) plus a simpler interpretable benchmark model for comparison and documentation.
- C. Use only a logistic regression regardless of accuracy loss.
- D. Use a black-box model and refuse to explain.

Answer: B

Explanation

Regulated environments commonly combine high-performance models with post-hoc explainability and interpretable benchmarks. Options A and D are non-compliant. Option C sacrifices accuracy without an explanation framework.

Question

A data scientist standardizes features using the full dataset, including test data, and then splits into train/test. Another analyst says this is wrong. Which is the correct critique?

- A. It is fine because standardization is a linear transformation.
- B. It leaks test-set information (mean and standard deviation) into the training process, biasing performance estimates.
- C. Standardization should only be applied to categorical features.
- D. Standardization is unnecessary.

Answer: B

Explanation

Scaling parameters must be estimated on the training set only. Using the full dataset leaks test information. Options A, C, and D misstate the issue.

Question

A risk analyst wants to capture the relationship between interest rates and bond returns, which is known to be non-linear (convexity). Which modeling approach is most appropriate?

- A. A linear regression of returns on rates.
- B. A model that explicitly allows for non-linearity, such as a polynomial regression, spline, tree-based model, or neural network.
- C. A logistic regression.
- D. A KNN classifier.

Answer: B

Explanation

Non-linear relationships require models that can capture curvature: polynomial regressions, splines, trees, or neural networks. Options A, C, and D are inappropriate.

Question

A research team fits a deep neural network to only 500 observations with 100 features. Test performance is poor. What is the most likely explanation, and what should be done?

- A. The model is too simple; increase depth.
- B. Overfitting due to small sample and high dimensionality; use simpler models, regularization, dimensionality reduction, or gather more data.
- C. The test set is corrupted.
- D. The target is mislabelled.

Answer: B

Explanation

With 500 observations and 100 features, a deep network will overfit. Remedies: simpler models, regularization, PCA, more data. Options A, C, and D are unlikely explanations.

Question

A time-series dataset of daily FX rates has occasional gaps (weekends, holidays, occasional missing days). Which imputation approach is generally most appropriate?

- A. Drop all rows with missing values.
- B. Forward-fill, interpolation, or model-based imputation that respects temporal structure.
- C. Replace with zeros.
- D. Replace with the global mean.

Answer: B

Explanation

Time-series missingness should be handled with methods that respect temporal order (forward-fill, interpolation, state-space models). Options A, C, and D distort the series.

Question

A dataset has a high-cardinality categorical variable “merchantid” with 10,000 unique values. One-hot encoding would produce 10,000 new columns. What is a better approach?

- A. Use one-hot encoding anyway.
- B. Use target encoding (with proper cross-validation to avoid leakage), hashing, or embedding-based representations.
- C. Remove the feature.
- D. Convert to ordinal codes.

Answer: B

Explanation

High-cardinality categoricals should use target encoding (with regularization), hashing, or embeddings. Option A causes dimension explosion; C loses information; D implies an order that does not exist.

Question

A portfolio returns dataset has a few extreme observations due to flash-crash-type events. Two models are being compared: OLS regression and quantile regression. Which is generally more robust to these extremes?

- A. OLS, because it uses all information.
- B. Quantile regression, because it targets conditional quantiles rather than the mean and is therefore less influenced by extreme observations.
- C. Both are equally robust.
- D. Neither; only tree models work.

Answer: B**Explanation**

Quantile regression estimates conditional quantiles, providing robustness to extreme values. OLS is sensitive. Options A, C, and D are incorrect.

Question

A team deploys a production ML pipeline. Which of the following testing categories is essential to ensure reliability?

- A. Unit tests on transformation functions.
- B. Integration tests on the full pipeline.
- C. Adversarial and stress tests on model behaviour.
- D. All of the above.

Answer: D

Explanation

Production ML pipelines need unit, integration, and adversarial/stress testing. All three are essential.

Question

A team reduces 300 candidate features to 25 using LASSO, then fits a logistic regression on the selected 25. Is this valid?

- A. Yes, this is standard and correct.
- B. It can be valid but has a subtle issue: feature selection using the full data introduces selection bias. Ideally, LASSO selection should be done inside each cross-validation fold to avoid leakage.
- C. No, LASSO should not be used for feature selection.
- D. No, logistic regression should be used directly.

Answer: B

Explanation

Selection on the full data leaks information. Nested CV avoids this by doing selection within each fold. Options A, C, and D misstate the issue or the role of LASSO.

Question

A K-means clustering is run on customer data with features: annual spend (USD 100–1M), tenure (months, 1–240), and number of products (1–10). The result is dominated by annual spend. What should the analyst do?

- A. Remove annual spend.
- B. Standardize or normalize the features before running K-means.
- C. Use fewer observations.
- D. Increase K.

Answer: B

Explanation

K-means uses distances; features on different scales dominate. Scaling fixes this. Options A, C, and D do not address the scale issue.

Question

A clustering with $K=4$ has average silhouette score 0.32. With $K=5$, the score is 0.15. What do these scores indicate?

- A. $K=5$ is better because the score is lower.
- B. $K=4$ is preferred because the average silhouette score is higher, indicating better-defined clusters.
- C. Neither is good.
- D. Silhouette scores only apply to $K=2$.

Answer: B

Explanation

Higher silhouette = better-separated clusters. $K=4$ is preferred. Options A, C, and D misstate the interpretation.

Question

A dendrogram on 500 customers shows two very long vertical branches near the top, then progressively shorter merges. How many clusters are suggested?

- A. 500 (each customer is its own cluster).
- B. 2 (the two branches separated by the long vertical lines).
- C. 10 (based on the number of intermediate branches).
- D. 1 (all merged).

Answer: B

Explanation

Long vertical lines indicate significant separation. Cutting above the long vertical lines yields 2 clusters. Options A, C, and D misread the dendrogram.

Question

A quant team is deciding between random forest and gradient-boosted trees for a medium-sized tabular classification problem (10,000 rows, 40 features). Which is generally true?

- A. Random forest is always better.
- B. Both are competitive; gradient boosting often achieves slightly higher accuracy with careful tuning, while random forest is generally more robust to hyperparameter choices and slightly faster to train.
- C. Gradient boosting is always better.
- D. Neither is suitable.

Answer: B

Explanation

Both are strong; boosting often wins on accuracy with tuning, while RF is easier to use. Options A, C, and D are overstated.

Question

An SVM with a linear kernel underperforms; with an RBF kernel it performs better. What does this suggest about the data?

- A. The classes are linearly separable.
- B. The classes are not linearly separable, and a non-linear decision boundary is required.
- C. The data is noisy.
- D. The SVM is broken.

Answer: B

Explanation

The improved performance with a non-linear kernel indicates the boundary is non-linear. Options A, C, and D are not supported.

Question

KNN performs poorly on a dataset with 200 features and 1,000 observations. What is a likely cause and remedy?

- A. Too many observations; reduce.
- B. Curse of dimensionality; use dimensionality reduction (PCA, feature selection) or a different algorithm.
- C. Wrong distance metric; use Manhattan.
- D. Increase k .

Answer: B**Explanation**

High-dimensional spaces suffer from distance concentration, making KNN unreliable. Dimensionality reduction or alternative algorithms are recommended. Options A, C, and D are secondary.

Question

A fully grown decision tree achieves 100% accuracy on training data and poor accuracy on test data. What should be done?

- A. Grow the tree deeper.
- B. Prune the tree (or apply pre-pruning constraints on depth, minimum samples per leaf, etc.) to reduce overfitting.
- C. Use a different dataset.
- D. Remove all features.

Answer: B

Explanation

Unpruned trees overfit; pruning reduces variance. Options A, C, and D do not address overfitting.

Question

An ensemble of 50 identical decision trees performs no better than a single tree. What is the explanation?

- A. Ensembles always improve performance.
- B. Diversity is essential: identical trees give identical predictions, so the ensemble has no variance reduction benefit. Bagging and feature randomness introduce the needed diversity.
- C. The trees are too shallow.
- D. The ensemble is too small.

Answer: B

Explanation

Ensembles improve performance when base learners are diverse and individually reasonable. Identical trees defeat the purpose. Options A, C, and D miss the point.

Question

A Naive Bayes classifier performs surprisingly well on text despite the conditional independence assumption being violated. Why?

- A. The assumption is satisfied in text data.
- B. Classification depends only on ranking class posteriors, not on exact probability calibration; Naive Bayes often ranks correctly even when probabilities are miscalibrated.
- C. Text data has no dependencies.
- D. Naive Bayes ignores the assumption in practice.

Answer: B

Explanation

Naive Bayes often ranks classes correctly even with miscalibrated probabilities because the decision boundary depends on the maximum, not on calibration. Options A, C, and D are wrong.

Question

A binary classification problem has features that are approximately normally distributed within each class with equal covariance matrices. Which is theoretically preferable: logistic regression or linear discriminant analysis?

- A. Logistic regression, always.
- B. LDA, because when its assumptions (normal features, equal covariance) hold, it is the optimal Bayes classifier.
- C. Both are equally good regardless of assumptions.
- D. SVM.

Answer: B

Explanation

Under normality and equal covariance, LDA is the Bayes-optimal classifier. Logistic regression is more robust to assumption violations and often used when these don't hold. Options A, C, and D are not correct.

Question

An analyst wants to predict whether a loan will default or not. Which type of model is appropriate?

- A. Linear regression.
- B. Logistic regression (or another classification model).
- C. Polynomial regression.
- D. Ridge regression.

Answer: B

Explanation

Default is a binary outcome; classification models are appropriate. Linear and polynomial regressions predict continuous outputs; Ridge is regularized linear regression. Option B is correct.

Question

Which of the following is NOT an assumption of standard Linear Discriminant Analysis?

- A. Features are normally distributed within each class.
- B. Classes share a common covariance matrix.
- C. Features are continuous.
- D. Classes have equal prior probabilities.

Answer: D

Explanation

LDA allows for unequal priors (estimated from data or specified). The other three assumptions are standard. Option D is not required.

Question

Two models are compared on a fraud dataset with 1% positives. Model A has AUC 0.90; Model B has AUC 0.75. But Model B has a higher precision-recall AUC (PR-AUC). Which model is better?

- A. Model A, because ROC AUC is always the criterion.
- B. Depends on the use case; PR-AUC is often more informative for highly imbalanced data, so Model B might be preferable if precision at high recall matters.
- C. Model B, always.
- D. Neither; discard both.

Answer: B

Explanation

For imbalanced data, PR-AUC often better reflects performance on the minority class. Choice depends on business context. Options A, C, and D are overstated.

Question

A bank has a default prediction model. Cost of approving a defaulter (FP) is \$50k; cost of rejecting a good borrower (FN) is \$5k. How should the threshold be chosen?

- A. Threshold = 0.5 always.
- B. Threshold should be higher than 0.5 to reduce false positives, since FP cost $\gg$ FN cost.
- C. Threshold should be lower than 0.5 to increase recall.
- D. Threshold choice is irrelevant.

Answer: B

Explanation

When FP is much costlier than FN, raising the threshold reduces FPs. Options A, C, and D misapply the cost matrix.

Question

Model A has precision 0.9, recall 0.2. Model B has precision 0.4, recall 0.9. Which model is likely appropriate for a fraud-detection system where missing fraud is very costly?

- A. Model A, because precision is high.
- B. Model B, because recall is high, catching more actual frauds.
- C. Both are equivalent.
- D. Neither.

Answer: B

Explanation

When missing positives is very costly, recall is prioritised. Model B's high recall is preferable. Options A, C, and D are less appropriate.

Question

A confusion matrix shows $TP=80$, $FP=20$, $FN=40$, $TN=860$. Compute the F1 score.

- A. 0.615.
- B. 0.727.
- C. 0.842.
- D. 0.500.

Answer: B

Explanation

Precision 0.80; Options A, C, and D miscompute.

Question

A team has 50 features and 200 observations. They expect some features to be irrelevant. Which regularization is most appropriate?

- A. L2 (Ridge).
- B. L1 (LASSO).
- C. Elastic Net.
- D. None.

Answer: B

Explanation

When many features may be irrelevant, LASSO's feature selection is valuable. Elastic Net is also reasonable if features are correlated. Options A and D are less appropriate.

Question

With only 100 observations, why might leave-one-out cross-validation (LOOCV) be preferred over k-fold?

- A. Because it is faster.
- B. Because it uses almost all data for training in each fold, reducing bias in the performance estimate when the sample is very small.
- C. Because it is more random.
- D. Because it avoids overfitting.

Answer: B

Explanation

LOOCV maximises training data per fold; useful for very small samples. It is computationally expensive and higher variance, but lower bias. Options A, C, and D misstate the trade-off.

Question

Which is a key difference between bootstrapping and k-fold cross-validation for model evaluation?

- A. Bootstrapping samples with replacement; k-fold partitions the data without replacement.
- B. Bootstrapping is always better.
- C. k-fold can only be used for classification.
- D. They are identical.

Answer: A

Explanation

Bootstrapping uses sampling with replacement; k-fold partitions the data. Both estimate generalisation. Options B, C, and D misstate the distinction.

Question

Why is standardization recommended before applying L1 or L2 regularization?

- A. It is not recommended.
- B. Because regularization penalises coefficients, and coefficients on differently scaled features are not directly comparable; standardizing puts them on equal footing.
- C. It removes outliers.
- D. It speeds up training.

Answer: B

Explanation

Regularization penalises coefficient magnitude. If features are not on comparable scales, the penalty is unfairly applied. Options A, C, and D are not the reason.

Question

A hidden layer uses sigmoid activation. After many layers, gradients vanish and training stalls. Which alternative activation is most likely to help?

- A. Another sigmoid layer.
- B. ReLU (or its variants like Leaky ReLU, ELU).
- C. Softmax.
- D. Linear.

Answer: B

Explanation

ReLU avoids saturation for positive inputs and mitigates vanishing gradients. Softmax is for output; linear is not typically used in hidden layers. Options A, C, and D are less appropriate.

Question

Why is Xavier or He initialization used for neural network weights?

- A. To initialize weights to zero.
- B. To maintain variance of activations and gradients across layers, avoiding vanishing/exploding gradients.
- C. To speed up training at the cost of accuracy.
- D. To reduce model size.

Answer: B

Explanation

Xavier/He initialization scales weights based on layer sizes, preserving signal magnitude. Options A, C, and D are incorrect.

Question

Batch normalization improves deep network training primarily by:

- A. Reducing model complexity.
- B. Normalizing layer inputs, which stabilises training, mitigates internal covariate shift, and often allows higher learning rates.
- C. Eliminating the need for activation functions.
- D. Reducing the number of parameters.

Answer: B

Explanation

Batch norm normalizes intermediate activations, improving stability and convergence. Options A, C, and D are incorrect.

Question

Dropout is used during training. What happens at inference time?

- A. Dropout is applied again.
- B. Dropout is turned off; weights are scaled to account for the expected effect of dropout during training.
- C. Model is retrained without dropout.
- D. Model is discarded.

Answer: B

Explanation

Dropout is a training-time regularizer. At inference, dropout is disabled and weights are scaled (or the framework handles it). Options A, C, and D are incorrect.

Question

Early stopping in neural network training means:

- A. Stopping before training starts.
- B. Monitoring validation error during training and stopping when it stops improving, preventing overfitting.
- C. Stopping after a fixed number of epochs regardless of performance.
- D. Stopping when training error is zero.

Answer: B

Explanation

Early stopping is a form of regularization: training halts when validation performance degrades. Options A, C, and D are different or incorrect.

Question

Why do CNNs use shared weights (convolutional filters) across spatial positions?

- A. To reduce training time only.
- B. To exploit spatial invariance and dramatically reduce the number of parameters compared to fully connected layers.
- C. Because convolution is faster than matrix multiplication.
- D. To avoid using activation functions.

Answer: B

Explanation

Weight sharing reduces parameters and provides translational invariance. Options A, C, and D miss the architectural rationale.

Question

What is the primary purpose of pooling layers in CNNs?

- A. To increase spatial resolution.
- B. To reduce spatial dimensions (downsampling), control overfitting, and provide translation invariance.
- C. To add parameters.
- D. To replace convolution.

Answer: B

Explanation

Pooling reduces spatial size and provides robustness to small translations. Options A, C, and D are incorrect.

Question

In a transformer, self-attention computes pairwise interactions between tokens. What is the primary benefit?

- A. Faster training on small data.
- B. Modelling long-range dependencies and contextual relationships regardless of token distance.
- C. Reducing parameter count.
- D. Handling images only.

Answer: B

Explanation

Self-attention captures relationships between tokens regardless of distance, unlike RNNs. Options A, C, and D are incorrect.

Question

Why do transformers need positional encodings?

- A. Because transformers lack recurrence and therefore have no inherent notion of order.
- B. Because transformers are too slow without them.
- C. Because they help with image processing.
- D. Because they increase model size.

Answer: A

Explanation

Self-attention is permutation-invariant; positional encodings inject order information. Options B, C, and D are incorrect.

Question

A key advantage of transformers over RNNs is:

- A. Less computational cost.
- B. Parallelism over sequence positions and better handling of long-range dependencies.
- C. Better handling of small datasets.
- D. Simpler architecture.

Answer: B

Explanation

Transformers parallelise across tokens and capture long-range dependencies via attention. RNNs are sequential. Options A, C, and D are less accurate.

Question

Two groups have equal true positive rates but unequal false positive rates under a model. Which fairness metric does the model satisfy?

- A. Demographic parity.
- B. Equal opportunity.
- C. Predictive rate parity.
- D. None.

Answer: B

Explanation

Equal opportunity requires equal TPR across groups. The model satisfies that but violates equalized odds (which requires equal FPR too) and demographic parity. Options A, C, and D are not satisfied.

Question

An algorithm trained to predict job performance is disproportionately rejecting female candidates. Removing gender from features does not fix the issue. What is the most likely cause?

- A. Random noise.
- B. Proxy discrimination via correlated features (e.g., university, career gaps).
- C. The training set is too small.
- D. The target is wrong.

Answer: B**Explanation**

Protected attributes are often captured by correlated proxies. Options A, C, and D are secondary or incorrect.

Question

Which of the following cannot be simultaneously satisfied when base rates differ between groups?

- A. Accuracy and precision.
- B. Demographic parity, predictive rate parity, and equal opportunity.
- C. Recall and F1.
- D. ROC and AUC.

Answer: B

Explanation

The Kleinberg-Chouldechova impossibility result applies to these three fairness metrics. Options A, C, and D are not the impossibility case.

Question

A model trained on 20 years of hiring data favours male candidates. Which type of bias?

- A. Measurement bias.
- B. Historical bias.
- C. Sampling bias.
- D. Deployment bias.

Answer: B

Explanation

Historical bias reflects past discrimination encoded in data. Options A, C, and D are other sources.

Question

A model trained on data generated by its own past predictions can become increasingly extreme. What is this phenomenon?

- A. Random noise.
- B. Feedback loop or self-reinforcing bias.
- C. Overfitting.
- D. Underfitting.

Answer: B

Explanation

Feedback loops occur when model outputs influence future training data, amplifying biases. Options A, C, and D are different phenomena.

Question

A neural network's internal weights are not human-interpretable, but SHAP values explain individual predictions. What is this model?

- A. Interpretable and explainable.
- B. Explainable but not interpretable.
- C. Interpretable but not explainable.
- D. Neither.

Answer: B

Explanation

SHAP provides post-hoc explainability; the network's internals remain opaque. Options A, C, and D misstate.

Question

Which is an example of opacity by “specialised knowledge”?

- A. Being unaware a decision was made by an algorithm.
- B. Even with access to the code, only experts can interpret it.
- C. Hiding the algorithm.
- D. Using encrypted data.

Answer: B

Explanation

Specialised-knowledge opacity arises when understanding requires expertise. Option A is secrecy; C is confidentiality; D is unrelated.

Question

An operator accepts AI recommendations without independent verification. What is this?

- A. Confirmation bias.
- B. Automation bias.
- C. Selection bias.
- D. Survivorship bias.

Answer: B

Explanation

Automation bias is over-reliance on automated systems. Options A, C, and D are different biases.

Question

Which of the following are recognised sources of AI-related reputational risk?

- A. Privacy breaches, algorithmic bias, and lack of explainability.
- B. Only privacy.
- C. Only financial losses.
- D. Only regulatory fines.

Answer: A

Explanation

Privacy, bias, and explainability are the three primary sources of AI-related reputational risk. Options B, C, and D are too narrow.

Question

Which is a societal risk of GenAI?

- A. High compute cost.
- B. Misinformation at scale and deepfake-enabled fraud.
- C. Slow inference.
- D. Privacy only.

Answer: B

Explanation

GenAI's societal risks include misinformation and deepfakes. Options A and C are technical; D is too narrow.

Question

An AI system overrides a patient's stated preference for a treatment. Which principle is violated?

- A. Nonmaleficence.
- B. Beneficence.
- C. Autonomy.
- D. Explainability.

Answer: C

Explanation

Autonomy is the patient's right to decide. Options A, B, and D are distinct principles.

Question

Which is the difference between nonmaleficence and beneficence?

- A. Nonmaleficence: avoid harm; beneficence: actively do good.
- B. Both mean the same.
- C. Beneficence: avoid harm; nonmaleficence: actively do good.
- D. Neither applies to AI.

Answer: A

Explanation

Nonmaleficence = do no harm; beneficence = do good. Options B, C, and D misstate.

Question

Egalitarianism and meritocracy can conflict. What does this illustrate about justice in AI ethics?

- A. Justice is uncontroversial.
- B. Different distributive justice concepts can conflict, requiring trade-offs.
- C. Justice is a mathematical rule.
- D. Justice is irrelevant to AI.

Answer: B

Explanation

Justice concepts can conflict; choosing between them is a normative judgment. Options A, C, and D are incorrect.

Question

A robo-advisor nudges users toward certain products through manipulative design. Which principle does this threaten?

- A. Autonomy.
- B. Justice.
- C. Nonmaleficence.
- D. Explainability.

Answer: A

Explanation

Manipulative design undermines autonomy. Options B, C, and D are different principles.

Question

Which of the following is an example of AI-driven manipulation?

- A. A calculator computes sums.
- B. A recommendation engine leverages detailed user profiles to shape opinions or behaviour without user awareness.
- C. A chatbot answers FAQs.
- D. An OCR system reads text.

Answer: B

Explanation

Manipulation involves leveraging profiling to influence users. Options A, C, and D are not manipulative.

Question

Which of the following is a key concern in AI existential risk discussions?

- A. Model accuracy.
- B. Superintelligence and the alignment problem.
- C. Data storage costs.
- D. IT infrastructure.

Answer: B

Explanation

Existential risk focuses on superintelligence and alignment with human values. Options A, C, and D are operational.

Question

What is a concern regarding AI and global inequality?

- A. AI is equally accessible everywhere.
- B. Countries with advanced AI capabilities may pull ahead economically, widening global inequality.
- C. AI reduces inequality.
- D. AI is irrelevant to inequality.

Answer: B

Explanation

Concentration of AI capability can widen global economic and political gaps. Options A, C, and D are incorrect or unsupported.

Question

Why is AI particularly relevant to misinformation campaigns?

- A. AI cannot generate text.
- B. AI can generate realistic content at scale cheaply, amplifying misinformation and disinformation.
- C. AI only produces accurate information.
- D. AI is too slow to matter.

Answer: B

Explanation

Generative AI makes content creation cheap and scalable, increasing misinformation risk. Options A, C, and D are incorrect.

Question

Which is a privacy concern associated with AI?

- A. AI improves accuracy only.
- B. AI systems can infer sensitive attributes or habits from seemingly innocuous data, potentially enabling surveillance.
- C. AI always protects privacy.
- D. AI has no privacy implications.

Answer: B

Explanation

AI can infer sensitive information from data, raising privacy and surveillance risks. Options A, C, and D are incorrect.

Question

Which of the following is a functional risk of GenAI?

- A. Hallucination of plausible but incorrect outputs.
- B. Reduced model size.
- C. Faster inference.
- D. Lower training cost.

Answer: A

Explanation

Hallucination is a functional risk of GenAI. Options B, C, and D are benefits, not risks.

Question

A self-driving car algorithm is tuned to minimise total harm, potentially at the cost of the passenger. Which ethical framework best supports this?

- A. Deontology.
- B. Virtue ethics.
- C. Utilitarianism.
- D. Divine command theory.

Answer: C

Explanation

Utilitarianism maximises aggregate welfare, potentially overriding individual rights. Options A, B, and D are different frameworks.

Question

A company refuses to use AI for certain tasks even if it would improve aggregate outcomes, because the tasks violate fundamental rights. Which ethical framework guides this?

- A. Utilitarianism.
- B. Deontology.
- C. Virtue ethics.
- D. Pragmatism.

Answer: B

Explanation

Deontology prioritises duties and rights over consequences. Options A, C, and D are different.

Question

What question would a virtue ethicist ask when designing an AI system?

- A. What maximises utility?
- B. What would a virtuous person/organisation do?
- C. What does the law say?
- D. What is cheapest?

Answer: B

Explanation

Virtue ethics focuses on character and virtuous behaviour. Options A, C, and D are other frameworks or criteria.

Question

Which of the following is essential for ethical data collection under GDPR?

- A. No notice.
- B. Explicit consent (in many cases) or another lawful basis, data minimisation, and transparent purpose specification.
- C. Unlimited retention.
- D. Sharing with third parties.

Answer: B

Explanation

GDPR requires lawful basis, minimisation, and transparency. Options A, C, and D violate GDPR.

Question

Which of the following best captures the GDPR right to be forgotten?

- A. Right to access data.
- B. Right to request erasure of personal data under certain conditions.
- C. Right to sell data.
- D. Right to receive compensation for processing.

Answer: B

Explanation

The right to erasure (Art. 17) allows deletion under specified conditions. Options A, C, and D are other rights.

Question

What does contextual integrity require?

- A. Data can be used freely once collected.
- B. Data use must respect the norms of the context in which it was provided.
- C. Data should be encrypted.
- D. Data should be sold.

Answer: B

Explanation

Contextual integrity (Nissenbaum) requires data use to align with context-specific norms. Options A, C, and D misstate or miss the concept.

Question

Which of the following is explicitly prohibited by the EU AI Act?

- A. Fraud detection.
- B. Social scoring and manipulative AI exploiting vulnerabilities.
- C. Credit scoring.
- D. Customer service chatbots.

Answer: B

Explanation

Social scoring, manipulative AI, and exploitation of vulnerabilities are prohibited. Other uses are regulated but permitted.

Question

A US firm serves EU customers. Does GDPR apply?

- A. No.
- B. Yes, GDPR applies extraterritorially when serving EU data subjects.
- C. Only if the firm has an EU office.
- D. Only if the customer complains.

Answer: B

Explanation

GDPR Art. 3 provides extraterritorial scope. Options A, C, and D are incorrect.

Question

The NIST AI RMF functions are:

- A. Train, Test, Deploy, Monitor.
- B. Govern, Map, Measure, Manage.
- C. Design, Build, Audit, Retire.
- D. Plan, Do, Check, Act.

Answer: B

Explanation

NIST AI RMF functions: Govern, Map, Measure, Manage. Options A, C, and D are other frameworks.

Question

Kant's categorical imperative asks us to:

- A. Maximise utility.
- B. Act only on maxims that could be universal laws.
- C. Cultivate virtue.
- D. Follow the sovereign.

Answer: B

Explanation

Kant's imperative demands universalizability. Options A, C, and D are other traditions.

Question

Which statement correctly distinguishes the three main ethical frameworks?

- A. All three are identical.
- B. Consequentialism focuses on outcomes; deontology on duties; virtue ethics on character.
- C. All three focus on outcomes.
- D. All three ignore outcomes.

Answer: B

Explanation

The three frameworks differ in focus: consequences, duties, character. Options A, C, and D are incorrect.

Question

Which of the following is NOT typically considered a privacy best practice?

- A. Data minimisation.
- B. Right to be forgotten.
- C. Contextual integrity.
- D. Unlimited data retention.

Answer: D

Explanation

Unlimited retention is contrary to privacy best practices. Options A, B, and C are established practices.

Question

Which of the following best describes a trade-off when using synthetic data?

- A. It is always superior to real data.
- B. It reduces privacy risk and can be designed to remove bias, but may not capture the full complexity of real data.
- C. It eliminates the need for validation.
- D. It is always cheaper to generate.

Answer: B

Explanation

Synthetic data reduces privacy risk but may lack realism. Options A, C, and D are overstatements.

Question

Why must an organisation make a normative choice among fairness metrics?

- A. Because all metrics agree.
- B. Because different metrics can conflict when base rates differ, so the choice reflects values and business context.
- C. Because fairness is a technical problem.
- D. Because regulators prescribe one metric.

Answer: B

Explanation

The impossibility result forces a choice among fairness criteria, reflecting normative considerations. Options A, C, and D are incorrect.

Question

Which is an example of local explainability?

- A. Feature importance for the whole model.
- B. SHAP values for a specific prediction.
- C. Partial dependence plot.
- D. Global surrogate model.

Answer: B

Explanation

SHAP for a single prediction is local. Options A, C, and D are typically global.

Question

A company wants to give actionable advice to rejected applicants. Which technique is most appropriate?

- A. Global feature importance.
- B. Counterfactual explanations (e.g., "If income were \$X higher, you would be approved").
- C. Model training accuracy.
- D. Model architecture description.

Answer: B

Explanation

Counterfactuals provide actionable insights. Options A, C, and D are not actionable for the individual.

Question

Why can't fairness be fully automated?

- A. It is too expensive.
- B. Because it involves normative choices among conflicting criteria, especially when base rates differ.
- C. Because there is not enough data.
- D. Because GDPR forbids it.

Answer: B

Explanation

Fairness involves value judgments. Options A, C, and D are not the fundamental reason.

Question

Which of the following is a legitimate concern about AI and global inequality?

- A. AI is naturally equitable.
- B. The concentration of AI capability in a few countries could worsen global disparities.
- C. AI is too slow to matter.
- D. AI reduces inequality automatically.

Answer: B

Explanation

Concentration of AI capability can widen global gaps. Options A, C, and D are incorrect or unsupported.

Question

Which best describes the privacy-utility trade-off in AI?

- A. More data always improves privacy.
- B. More data can improve model utility but increases privacy risk; techniques like differential privacy can help balance.
- C. Privacy and utility never conflict.
- D. Privacy is always more important.

Answer: B

Explanation

Data utility comes at privacy cost; techniques like differential privacy manage the trade-off. Options A, C, and D are oversimplifications.

Question

The primary benefit of an AI ethics framework is:

- A. Increasing model accuracy.
- B. Providing structured guidance for ethical decisions and reducing regulatory and reputational risk.
- C. Reducing training time.
- D. Eliminating all ethical issues.

Answer: B

Explanation

Ethics frameworks provide guidance and risk reduction. Options A, C, and D are not their primary purpose.

Question

Which of the following is NOT typically a component of a data governance framework?

- A. Data strategy and quality.
- B. Data provenance and classification.
- C. Metadata management.
- D. Marketing strategy.

Answer: D

Explanation

Data governance covers strategy, quality, provenance, classification, metadata, protection, access, and compliance. Marketing is separate.

Question

Data provenance refers to:

- A. The data storage location.
- B. The origin, history, and legal right to use data.
- C. The data transformation steps only.
- D. The data format.

Answer: B

Explanation

Provenance tracks origin, ownership, and legal rights. Options A, C, and D are partial or incorrect.

Question

Which of the following best describes data classification?

- A. Sorting data alphabetically.
- B. Categorising data by sensitivity and structure to determine handling and controls.
- C. Storing data in a database.
- D. Compressing data.

Answer: B

Explanation

Classification determines appropriate handling and protections. Options A, C, and D are unrelated.

Question

Metadata management is important because:

- A. Metadata replaces data.
- B. Metadata supports discovery, lineage, governance, and understanding of data assets.
- C. Metadata is encrypted data.
- D. Metadata is unnecessary.

Answer: B

Explanation

Metadata enables discovery, lineage, and governance. Options A, C, and D misstate or dismiss.

Question

Role-Based Access Control (RBAC) grants access based on:

- A. Individual discretion.
- B. User roles within the organisation.
- C. Random selection.
- D. Data size.

Answer: B

Explanation

RBAC grants access based on roles. Options A, C, and D are not RBAC.

Question

A comprehensive model governance framework should include:

- A. Definitions of models and non-models, risk tiering, validation standards, documentation, monitoring, and reporting.
- B. Only model training.
- C. Only model deployment.
- D. Only model retirement.

Answer: A

Explanation

Governance covers definitions, tiering, validation, documentation, monitoring, and reporting. Options B, C, and D are partial.

Question

The primary purpose of model documentation is to:

- A. Improve accuracy.
- B. Provide a record of design, assumptions, limitations, and usage to support validation, audit, and governance.
- C. Speed training.
- D. Reduce features.

Answer: B

Explanation

Documentation supports validation, audit, and governance. Options A, C, and D are incorrect purposes.

Question

A model inventory should typically include:

- A. Model owner, purpose, risk tier, validation status, and interdependencies.
- B. Only model names.
- C. Only model versions.
- D. Only model code.

Answer: A

Explanation

A model inventory is comprehensive: owner, purpose, risk tier, validation, and interdependencies. Options B, C, and D are incomplete.

Question

A model landscape differs from a model inventory in that it:

- A. Lists all models.
- B. Shows interactions and interdependencies between models, highlighting cascade risks.
- C. Lists model code.
- D. Replaces the inventory.

Answer: B

Explanation

Landscape shows interactions; inventory is a registry. Options A, C, and D are incorrect.

Question

In the Three Lines of Defence model for AI/ML, who is primarily responsible for independent validation?

- A. First line (model developers).
- B. Second line (independent risk management/model validation).
- C. Third line (internal audit).
- D. External auditors.

Answer: B

Explanation

The second line performs independent validation. First line owns; third line audits; external auditors add extra assurance. Options A, C, and D misstate the roles.